

Carrigé Type Structure 2017

Exercise 1

$$X = 63.0265 = (111111,0001)_2 = (1,111110001)_2 \times 2^{-5} / ED = ER + 127 = 5 + 127 = +132$$

$$X = (0\ 10000100\ 111110001000000000000000)_2 = (427C4000)_{16}$$

$$X = -96.75 = -(1100000,11)_2 = -(1,10000011)_2 \times 2^{-6} / ED = ER + 127 = 6 + 127 = +133$$

$$X = (1\ 10000101\ 100000110000000000000000)_2 = (C2C18000)_{16}$$

$$X = +28.8 = (11100,11001100...) = (1,11001100...) \times 2^{-4} / ED = ER + 127 = 4 + 127 = +131$$

$$X = (0\ 10000011\ 11001100110011001100110)_2 = (41E66666)_{16}$$

$$(C020800000000000)_{16} = -8.25$$

$$(4020800000000000)_{16} = +8.25$$

$$(7FF0000000000000)_{16} = +\text{inf}$$

$$(FFFA000000000000)_{16} = \text{NaN}$$

Exercise 2

$$\begin{aligned} 1. & AB + A\bar{B}\bar{C} + BC + \bar{A}B\bar{C} + AC + \bar{A}\bar{B}C \\ &= A(B + \bar{B}\bar{C}) + B(C + \bar{A}\bar{C}) + C(A + \bar{A}\bar{B}) \\ &= A((B + \bar{B})(B + \bar{C})) + B((C + \bar{A})(C + \bar{C})) + C((A + \bar{A})(A + \bar{B})) \\ &= A(B + \bar{C}) + B(C + \bar{A}) + C(A + \bar{B}) \\ &= AB + B\bar{A} + BC + \bar{B}C + A\bar{C} + AC \\ &= A(\bar{C} + C) + B(\bar{A} + A) + C(\bar{B} + B) \\ &= A + B + C \end{aligned}$$

$$2. A + \bar{A}(\bar{B}\bar{C}\bar{D} + C + D) + B\bar{D} = 1$$

$$3. A(\bar{B}\bar{C} + BC) + \bar{A}B\bar{C} + \bar{A}\bar{B}C = A \oplus B \oplus C \quad \text{autre solution correcte } A \oplus B \oplus C$$

Exercise 3

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	
11	1	1	1	
10				

$$= \bar{C}\bar{D} + \bar{A}.D + B.D$$

AB \ CD	00	01	11	10
00	1	1		1
01	1	1	1	
11		1	1	
10	1			1

$$= \bar{B}.D + B.D + \bar{A}.\bar{C}$$

AB \ CD	00	01	11	10
00	1			1
01	1			1
11	1	1	1	1
10		1	1	

$$= B.C + \bar{B}.\bar{C} + C.D$$

$$= B.C + \bar{B}.\bar{C} + \bar{B}.D$$

Corrigé Type Structure 2017

Exercice 2

A	B	C	F(A,B,C)
0	0	0	0
1	0	0	0
0	1	0	1
1	1	0	1
0	0	1	1
1	0	1	0
0	1	1	1
1	1	1	0

$$F(A,B,C) = \sum(2,3,4,6)$$

$$F(A,B,C) = \prod(0,1,5,7)$$

Exercice 3

1. $F(A,B,C,D) = (\overline{A \cdot C} + \overline{A \cdot C})(B \oplus D)$

2. $F(A,B,C,D) = (A \oplus C)(B \oplus D)$

